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Real-Time Sitting Posture Detection for Students in Classroom Environment

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***Abstract*— Discomfort in the lower back, neck, and almost everywhere in the body can easily be noted among several people, particularly among students who spend long hours seated in classrooms. Extended periods of sitting in fixed postures can lead to strain and musculoskeletal issues, affecting focus and overall well-being. Real-time posture detection is necessary to prevent long-term health problems caused by poor posture and lifestyles, such as extended periods of sitting and low levels of physical activity. This paper proposes a system for real-time detection of students' sitting postures using the MediaPipe framework, specifically employing the angle heuristics approach. MediaPipe BlazePose is able to detect up to 33 keypoints, compared to the detection of fewer keypoints with most standard approaches. The system can recognize common postures like forward leaning, backward leaning, slouching, or crossing arms/legs.**

Keywords-Posture Detection, MediaPipe, Blazepose, Keypoints

I. INTRODUCTION

With the rise of digital age, students are spending their more time sitting at their computers or using electronic devices for learning due to this poor posture has become a widespread concern. The whirlwind shift to online education and increased usage of digital devices for educational purposes has resulted in prolonged sitting, mostly in inappropriate positions, leading to various health disorders, particularly musculoskeletal disorders [1]. Poor sitting postures, which students are too naive to know the lurking consequence in later life, can create chronic back pain, neck, and shoulder pain. At the same time, improper posture results in fatigue, loss of focus, and general discomfort in life that disrupts physical well-being and performance at school.

Research into similar health monitoring systems highlights the importance of conducting feasibility studies to assess the practicality of such interventions [2]. Feasibility studies have shown that real-time feedback systems can effectively encourage corrective posture habits, improving ergonomics and reducing health risks over time. Such systems continuously monitor posture, enabling students to maintain proper alignment. They can detect, in real time, common posture issues such as leaning forward, leaning backward, and slouching. This paper proposes a real-time posture detection system specifically designed for students, utilizing the MediaPipe framework. MediaPipe offers robust, real-time tracking of body's keypoints, enabling the detection of various postures [3]. The Blazepose model for pose estimation detects 33 key body landmarks such as shoulders, elbows, wrists, hips, knees, and ankles [4]. The key objective of this model is to identify common problems of posture that generally occurs due to extended periods of sitting in inappropriate position. The immediate feedback and guidance are supposed to lead to an improvement in the general ergonomics within the study environment of the students, helping them create healthy learning habits and avoiding posture-related health issues [5].

Key contributions of this paper are summarized as follows:

- Threshold-based classification is carried out from the geometric analysis of the postures. Six sitting postures are considered which are Leaning Forward, Leaning Backward, Slouching, Upright, Folded Arms and Crossed Legs.

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- Estimation of postures from the real-time input is presented.

The paper is further divided into the following sections:

Section 2 gives an overview of the related work Activity. Section 3 is the description of methodology taken to implement the posture detection in detail. Section 4 would be the result analysis. Section 5 would outline the conclusion and future orientation.

II. Related work

Several works related to posture detection have appeared previously but these have utilized complicated deep learning techniques including CNN and LSTM to recognise human behaviours in real time [6-8]. Additionally, Keypoints detection models like OpenPose and BlazePose have also become very common for the analysis of real-time body joint and limb positioning and, therefore, enable high levels of posture analysis. Due to their flexibility, they have been applied to many domains ranging from health to sports, ergonomic evaluations, even in the education field with an aim of further enhancing real-time posture correction and human activity recognition.

Kotak D. et al implemented yoga pose estimation using angle heuristic approach [9]. Firstly, 33 body keypoints have been identified and extracted and after that using angle heuristic method yoga poses were classified.

Weili D. et al proposed a method of pose recognition using the skeletal data gathered by a Kinect camera [10]. The new algorithm utilizing multiple features and rule-based learning, is capable of recognizing various human postures. The rule-based approach improves interpretability, leading to increase accuracy for posture recognization.

The work of Jiawei H. and Ding Z. provided novel insights into the area of classroom posture classification by proposing a hybrid computer vision methodology using advanced Open VINO (Open Visual Inference & Neural Network Optimization) technology, which determined the optimum ways of detecting postures [11]. A surveillance camera was used to capture videos of classroom environments in real time and The classification accuracy reached 93.3% for standing, 77.2% for sitting, and 95.9% for hand-raising, respectively.

Yiwen Z. et al proposed a new approach about posture recognition of students in classrooms [12]. The research applied YOLOv3 that is retrained to sense slumped students and build a pose estimation network. The model has performed well: to the COCO validation set, it gives 73.76% precision, and in recognition on a custom classroom dataset, it shows 90.1%; it was accurate and robust.

According to Zewei D. et al the evaluation of postures shall be achieved by categorizing them into predefined classes [13]. This does not call for explicit measurement of the variables required in computing risk scores. Each posture class is connected with a specific upper body configuration and an ergonomics risk score assigned using scoring methods such as Rapid Upper Limb Assessment (RULA).

Lin F-C et al introduced a system for recognizing student behavior using skeleton pose estimation [14]. Frames from a classroom camera are processed with OpenPose to collect skeleton data, followed by an error correction method to improve accuracy. Key features like joint positions, distances, and bone angles are extracted for behavior classification. A deep neural network is used to identify student actions and count classroom occupants.

Huang X. et al introduced Wearable sensing systems have been widely explored for human motion and posture recognition, focusing on indicators such as trunk, joints, and limbs, with sensors strategically placed to monitor physical changes (Author et al., Year) [15]. While effective, these systems often require physical attachment, raising issues of comfort and maintenance, especially in continuous monitoring scenarios. Camera-based systems offer a powerful alternative by providing non-contact, real-time posture monitoring, ideal for settings like classrooms where unobtrusive, continuous tracking is essential.

III. Methodology

The proposed method utilizes MediaPipe along with the BlazePose model. Common pre-trained pose detection architectures are used to analyse human postures. Applying pre-built pose detection models such as OpenPose, PoseNet, and DeepCut allows complex features, but these solutions often require high system resources and extensive data. Convolutional Neural Networks (CNNs), a common deep learning technique for computer vision tasks, also demand large volumes of data for effective training. In contrast, the angle-based approach for posture detection, utilizing MediaPipe avoids the need for extensive training data and high computational resources. The system calculates distances and angles between key points to determine specific postures based on side or front camera views. The specifics of MediaPipe and the selected method are explored in depth in the later sections.

A. Body Keypoints Extraction

As shown in Figure 1, MediaPipe Pose is a detailed body pose tracking solution that renders 33 3D landmarks and a background segmentation mask on the whole body from RGB video frames. The wide range of possible postures, coupled with slight changes in body positioning, obstruction of other objects can also affect the system's ability to consistently differentiate between postures. BlazePose ensures precise posture detection, making it ideal for monitoring. Refer to Figure 1 for the BlazePose 33 keypoints topology, which illustrates the specific landmarks detected by the model.

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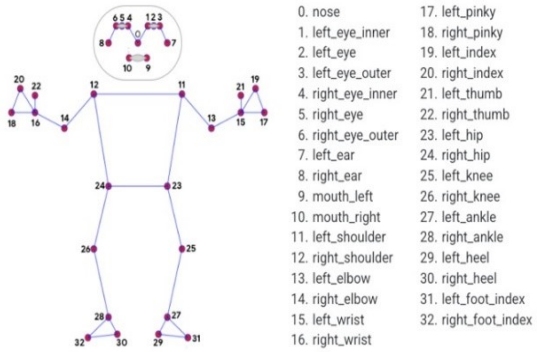


Fig. 1. BlazePose 33 Keypoints topology [18]

B. Angle Heuristic Approach

The angle heuristics methodology is used to classify Keypoints above the waist (shoulder, spine, neck) and below the waist (hip, knee, ankle) are extracted. Conditions are then applied based on specific angle thresholds for each posture, and if the posture meets the set conditions, the result is displayed, showing whether the posture is correct or incorrect. Similar conditions are applied for detecting different sitting (slouching, leaning forward-backword, cross leg etc.), providing real-time feedback to encourage proper posture.

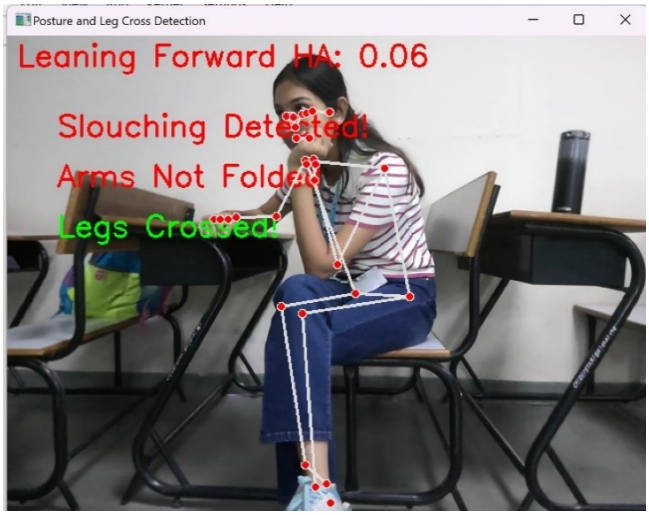


Fig. 2. Leaning Forward with leg crossed

The classification of the posture shown Figure 2 can be described based on the analysis of specific angle measurements, as discussed below.

- "Leaning Forward HA: 0.06": Horizontal alignment (difference between the shoulder midpoint and hip midpoint.) is positive(forward).
- “Slouching detected”: Angle between shoulder, hip and knee landmark is below the threshold (100 degree).
- Arms Folding: distance between left wrist a right elbow, and the right wrist and left elbow, are both greater that the 0.3 threshold.
- Legs Cross: The Position of the knee and ankle markers overlap.

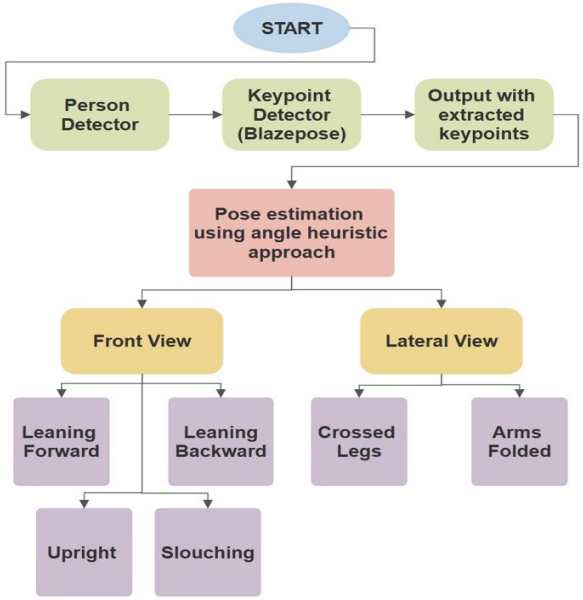


Fig. 3. Workflow of proposed work

Figure 3 outlines the process flow of the system. First, input images are manually collected and used as the main data source. These images are then fed into a machine learning pipeline using MediaPipe. During the pose detection process, the system identifies and extracts 33 key body points which includes critical joints like shoulders, elbows, hips, and knees, which help in understanding overall body posture.

Key body landmarks are used to compute joint angles mathematically. For each posture classification, the angle between three critical points (shoulder, hip, and knee) is calculated using the following formula:

$$\theta=\arctan2(c2-b2,c1-b1)-\arctan2(a2-b2,a1-b1)$$

i. Leaning:

For leaning posture detection, two primary positions are evaluated: leaning forward and leaning backward. The angle is computed using the vertical and horizontal alignment of the shoulder and hip landmarks calculated using midpoint of the shoulders and hips.

$$\theta = \arctan2(y_{hips} - y_{shoulders}, x_{hips} - x_{shoulders})$$

To classify the posture as either leaning forward or leaning backward, the system uses pre-defined thresholds.

- Forward Leaning: HA>0.
- Backward Leaning: HA<−0.05
- Upright: −0.05≤HA≤0.05

ii. Slouching:

To detect slouching, the angle formed between the line segments connecting the shoulder to the hip and the hip to the knee. This approach helps to assess the curvature of the back.

$$\theta = \arctan2(y_{knee} - y_{hip}, x_{knee} - x_{hip}) - \arctan2(y_{shoulder} - y_{hip}, x_{shoulder} - x_{hip})$$

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The threshold is set to 100 degrees for optimal results, ensuring accurate detection of slouching by effectively differentiating between upright and slouched postures.

iii. Arms Folded:

It measures the distances between the left wrist and right elbow, and vice versa. If both distances fall below a predefined threshold, it signifies that the arms are folded.

- Distance Between Wrists and Opposite Elbows:

$$D_{\text{left-wrist-right-elbow}} = \sqrt{(x_{\text{left-wrist}} - x_{\text{right-elbow}})^2 + (y_{\text{left-wrist}} - y_{\text{right-elbow}})^2}$$

$$D_{\text{right-wrist-left-elbow}} = \sqrt{(x_{\text{right-wrist}} - x_{\text{left-elbow}})^2 + (y_{\text{right-wrist}} - y_{\text{left-elbow}})^2}$$

- Threshold for arms folded:

$$D_{\text{left-wrist-right-elbow}} < 0.3 \text{ and}$$

$$D_{\text{right-wrist-left-elbow}} < 0.3$$

iv. Crossed Legs:

Measure the distance between opposite knees and ankles. If the knee distance is below a certain threshold and the ankle distance is above a threshold, the legs are considered crossed.

$$d_{\text{knees}} = \sqrt{(P_{LKx} - P_{RKx})^2 + (P_{LKy} - P_{RKy})^2}$$

$$d_{\text{ankles}} = \sqrt{(P_{LAx} - P_{RAx})^2 + (P_{LAy} - P_{RAy})^2}$$

IV. Result and Analysis

The section reports the results of our experimental work. The experiments used a system with an Intel(R) Iris(R) Xe 8GB GPU, 16GB RAM, a 13th Gen Intel(R) Core i7 CPU, and a 512GB SSD. This system, depending on BlazePose, would identify 33 key points on the human body, causing it to eventually detect common posture-related problems such as forward leaning, backward leaning, and slouching. Continuous Camera Observation and angle heuristic approach used to detect postures. During testing, the system showed robustness under various conditions, camera performance, conditions of lighting, and significant height difference among people. Poor lighting or low-resolution camera settings could lead to incorrect landmark detection, and therefore overall classification accuracy could be affected. The MediaPipe library excels in landmark prediction toward accurate posture classification for both males and females but excluded postures in the analysis will eventually yield less accurate results. The posture classifications and corresponding angle measurements are summarized in Table 1.

(a)

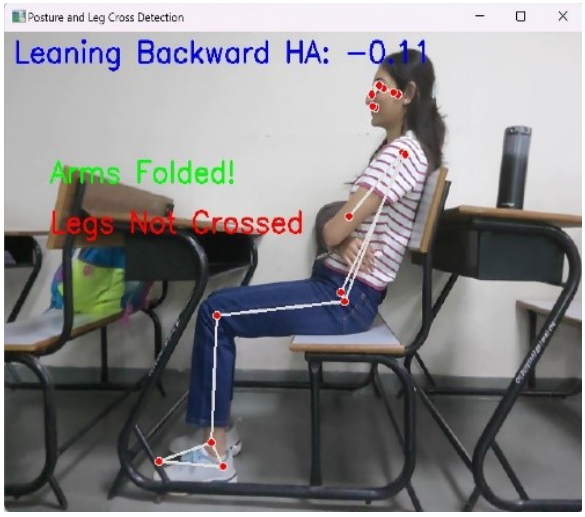


Fig. 4. Leaning Backward with Arms Folded

The system correctly identifies as leaning backward with arms folded as shown in Figure 4.
Leaning Backward: Horizontal alignment is negative.
Arms Folded: Arms not folded determined by the distance between the wrists and opposite elbows.
Legs not crossed: The Position of the knee and ankle markers does not overlap.

(b)



Fig. 5. Leaning Forward with Upright, Arms Folded and Legs Crossed

As shown in Figure 5, the system correctly identifies as leaning forward along with upright, arms folded and legs crossed.
Leaning Forward: Horizontal alignment is Positive.
Upright: Angle between shoulder-hip-knee could be $95° \leq \text{angle} \leq 110°$.
Arms Folded: Arms not folded determined by the distance between the wrists and opposite elbows could be <0.3.
Legs crossed: The Position of the knee and ankle markers overlap.

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(c)



Fig. 6. Leaning Forward with Arms Folded and Legs Crossed

As depicted in Figure 6, the system accurately identifies the posture as leaning forward with arms folded and legs crossed.
Leaning Forward: Horizontal alignment is Positive.
Arms Folded: The distance between the wrists and opposite elbows could be <0.3.
Legs crossed: The Position of the knee and ankle markers overlap.

(d)

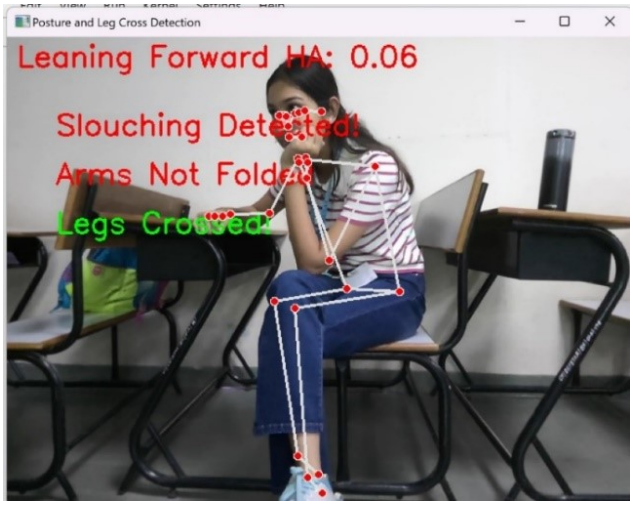


Fig. 7. Leaning Forward with Slouching

Figure 7 illustrates that the system correctly identifies the posture as leaning forward with slouching.
Leaning Forward: Horizontal alignment is Positive.
Slouching: Angle between shoulder-hip-knee could be <95°.
Arms not Folded: The distance between the wrists and opposite elbows is >0.3.
Legs crossed: The Position of the knee and ankle markers overlap.

(e)

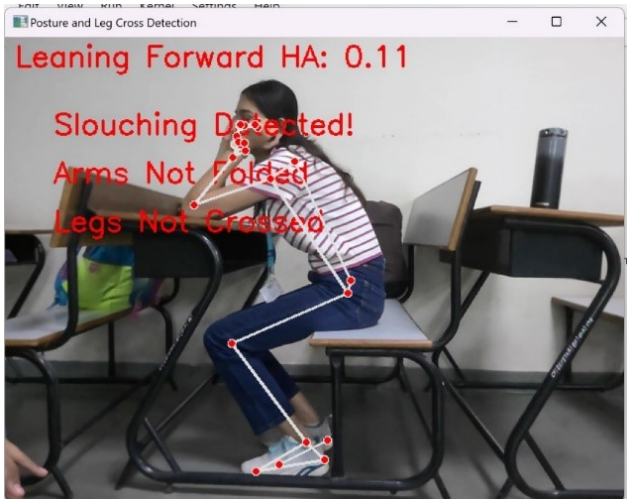


Fig. 8. Slouching, Arms not folded and legs not crossed

In Figure 8, the system successfully detects slouching, with arms not folded and legs uncrossed.
Leaning Forward: Horizontal alignment is Positive.
Slouching: Angle between shoulder-hip-knee could be <95°.
Arms not Folded: The distance between the wrists and opposite elbows is >0.3.Legs not crossed: The Position of the knee and ankle markers does not overlap.

(f)



Fig. 9. Upright Posture

As can be seen in Figure 9, the system is able to identify a person with Upright posture.
Leaning Forward: Horizontal alignment is Positive.
Upright: Angle between shoulder-hip-knee could be $95° \leq \text{angle} \leq 110°$.
Arms not Folded: The distance between the wrists and opposite elbows is >0.3.
Legs not crossed: The Position of the knee and ankle markers does not overlap.

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Table 1 Table of angles with posture status

Posture Detected	Back Angle (Shoulder-Hip-Knee)	Leaning Status	Arms Folded	Legs Crossed
Upright	95° ≤ angle ≤ 110°	Upright	No	No
Slouching	angle < 95°	Leaning Forward	No	No
Leaning Forward	-	Leaning Forward	No	No
Leaning Backward	-	Leaning Backward	No	No
Upright with Folded Arms	95° ≤ angle ≤ 110°	Upright	Yes	No
Slouching with Folded Arms	angle < 95°	Leaning Forward	Yes	No
Upright with Legs Crossed	95° ≤ angle ≤ 110°	Upright	No	Yes
Slouching with Legs Crossed	angle < 95°	Leaning Forward	No	Yes

V. Conclusion and Futurework

The article is a real-time posture detection system using the MediaPipe framework specifically designed for students who increasingly suffer from problems related to poor posture during long digital learning sessions. The angle heuristic strategy used in the system here successfully acts as a facilitator for real-time posture recognition, though it is very sensitive to very comparable postures that makes it unable to recognize sometimes. Limitations like these, based on more accurate angle definitions may further help its reliability in distinguishing between slight posture differences. The future work may be advanced deep learning, which could lead to the precise identification of keypoints as well as a wider range of detectable postures. More wearable devices can enhance adaptability for educational settings that promotes healthier long-term sitting habits.

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References

[1] Piñero-Fuentes, E.; Canas-Moreno, S.; Rios-Navarro, A.; Domínguez-Morales, M.; Sevillano, J.L.; Linares-Barranco, A. A Deep-Learning Based Posture Detection System for Preventing Telework-Related Musculoskeletal Disorders. *Sensors* **2021**, *21*, 5236. https://doi.org/10.3390/s21155236

[2] S. Liaqat, K. Dashtipour, K. Arshad, K. Assaleh and N. Ramzan, "A Hybrid Posture Detection Framework: Integrating Machine Learning and Deep Neural Networks," in *IEEE Sensors Journal*, vol. 21, no. 7, pp. 9515-9522, 1 April1, 2021, doi: 10.1109/JSEN.2021.3055898.

[3] Lin, Y. , Jiao, X. and Zhao, L. (2023) Detection of 3D Human Posture Based on Improved Mediapipe. *Journal of Computer and Communications*, **11**, 102-121. doi: 10.4236/jcc.2023.112008.

[4] Valentin Bazarevsky and Ivan Grishchenko et al.(2020). BlazePose: On-device Real-time Body Pose tracking https://doi.org/10.48550/arXiv.2006.10204

[5] Trabelsi, Z.; Alnajjar, F.; Parambil, M.M.A.; Gochoo, M.; Ali, L. Real-Time Attention Monitoring System for Classroom: A Deep Learning Approach for Student's Behavior Recognition. *Big Data Cogn. Comput.* **2023**, *7*, 48. https://doi.org/10.3390/bdcc7010048

[6] N. M. Alruwais and M. Zakariah, "Student Recognition and Activity Monitoring in E-Classes Using Deep Learning in Higher Education," in *IEEE Access*, vol. 12, pp. 66110-66128, 2024, doi: 10.1109/ACCESS.2024.3354981

[7] A. Kavitha, S. K, S. L G, V. J and V. N, "Framework for Detecting Student Behaviour (Nail Biting, Sleep, and Yawn) Using Deep Learning Algorithm," *2024 2nd International Conference on Artificial Intelligence and Machine Learning Applications Theme: Healthcare and Internet of Things (AIMLA)*, Namakkal, India, 2024, pp. 1-6, doi: 10.1109/AIMLA59606.2024.10531573.

[8] Lee M-FR, Chen Y-C, Tsai C-Y. Deep Learning-Based Human Body Posture Recognition and Tracking for Unmanned Aerial Vehicles. *Processes*. 2022; 10(11):2295. https://doi.org/10.3390/pr10112295

[9] D. Kotak, J. Desai, R. Shah, P. Goel and K. Patel, "Yoga Pose Classification using Angle Heuristic Approach," *2022 4thInternational Conference on Inventive Research in Computing Applications (ICIRCA)*, Coimbatore, India, 2022, pp. 1709-1714, doi: 10.1109/ICIRCA54612.2022.9985555.

[10] Ding, W., Hu, B., Liu, H. *et al.* Human posture recognition based on multiple features and rule learning. *Int. J. Mach. Learn. & Cyber.* **11**, 2529–2540 (2020). https://doi.org/10.1007/s13042-020-01138-y.

[11] Huang, J., Zhou, D. A scalable real-time computer vision system for student posture detection in smart classrooms. *Educ Inf Technol* **29**, 917–937 (2024). https://doi.org/10.1007/s10639-023-12365-5

[12] Zhang, Y., Zhu, T., Ning, H. *et al.* Classroom student posture recognition based on an improved high-resolution network. *J Wireless Com Network* **2021**, 140 (2021). https://doi.org/10.1186/s13638-021-02015-0

[13] Ding, Zewei & Li, Wanqing & Ogunbona, Philip & Qin, Ling. (2019). A real-time webcam-based method for assessing upper-body postures. Machine Vision and Applications. 30. 833-850. 10.1007/s00138-019-01033-9.

[14] Lin F-C, Ngo H-H, Dow C-R, Lam K-H, Le HL. Student Behavior Recognition System for the Classroom Environment Based on Skeleton Pose Estimation and Person Detection. *Sensors*. 2021; 21(16):5314. https://doi.org/10.3390/s21165314

[15] Huang X, Xue Y, Ren S, Wang F. Sensor-Based Wearable Systems for Monitoring Human Motion and Posture: A Review. *Sensors*. 2023; 23(22):9047. https://doi.org/10.3390/s23229047

[16] Mediapipe Library documentation https://ai.google.dev/edge/mediapipe/solutions/guide

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